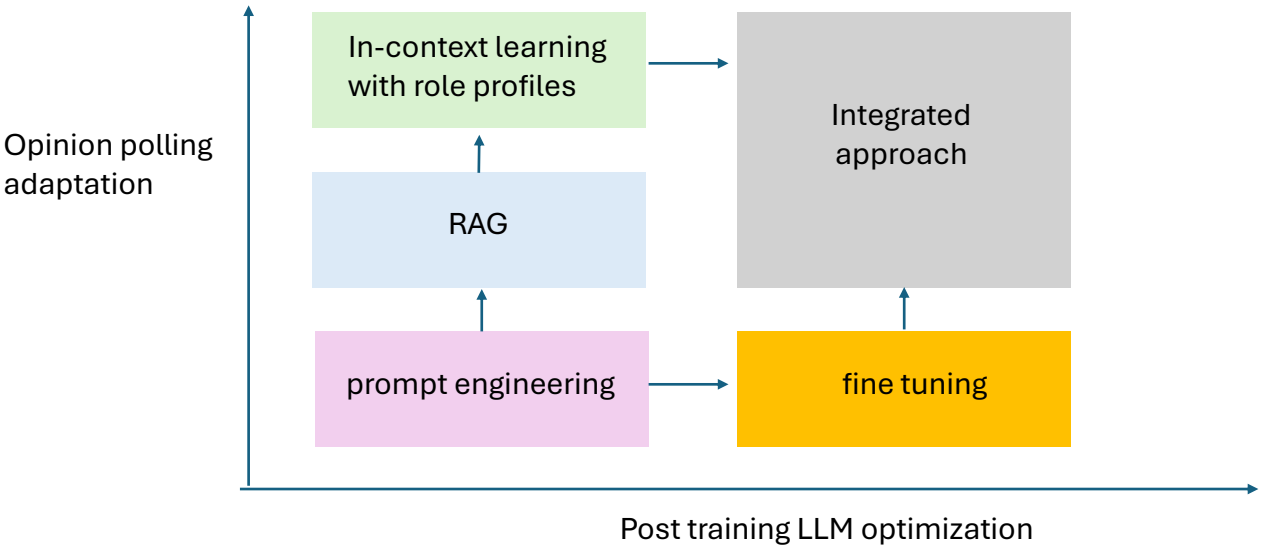

Figure 2 Adapting LLMs to synthesizing public opinion tasks.

fine-tuning incurs higher costs, requires AI expertise, and takes time. This method contrasts with other post-training approaches where the model weights are not changed.

The most straightforward post-training approach is prompt engineering, which includes zero-shot and few-shot learning. In the case of zero-shot learning, a user prompts an LLM without any examples, attempting to take advantage of the reasoning patterns it has gleaned in a general-purpose LLM. In zero-shot learning, a user provides an LLM with a prompt without any examples, aiming to leverage the reasoning patterns the model has acquired during its general-purpose training. Prompt engineering can enhance the performance of a pre-trained model; however, this improvement is often limited.

Another line of post-training optimization is knowledge injection Lauscher et al. (2020); Chen et al. (2022). By incorporating knowledge bases such as political affiliation, polling results, political ideology, and demographics, it is possible to teach a pre-trained model about the specific domain and tasks. Fine-tuning is one way to add knowledge to a pre-trained model. With fine-tuning, the model builder continues the model training process and adapts it using task-specific data. By exposing the model to a specific knowledge domain, its weights can be adapted for the targeted applications. As a result, its performance in particular tasks, such as polling simulation, can be more relevant to the specialized domains.

Another approach to improve a model's knowledge base is using in-context learning, such as retrieval augmented generation (RAG) Fan et al. (2024); Lewis et al. (2020). RAG utilizes information retrieval techniques to allow general-purpose LLMs to access relevant data from a knowledge source, often stored in vector databases, and integrate it into the generated text Jing et al. (2024). Post-training knowledge injection can address limitations for many knowledge-intensive tasks Ovadia et al. (2024). However, general post-training knowledge injections, such as existing RAG approaches, are not explicitly designed for role-play and opinion polling tasks.

A promising direction is RAG enhanced with role profiles. RAG—augmented with role profiles—can simulate the political opinions of specific population groups more accurately than simple RAG. We introduce this approach to the literature and present experimental results to demonstrate its advantages for polling tasks.

3 Simulating Voter Preferences

Several studies investigate the feasibility and effectiveness of using LLMs to simulate survey responses. A key study by Argyle et al. (2023) finds that LLMs can provide reasonably accurate simulations of group-level responses in behavioral science and economics experiments, as well as for political surveys. Horton (2023) and Aher et al. (2023) further corroborate these findings, demonstrating the potential of LLMs to capture

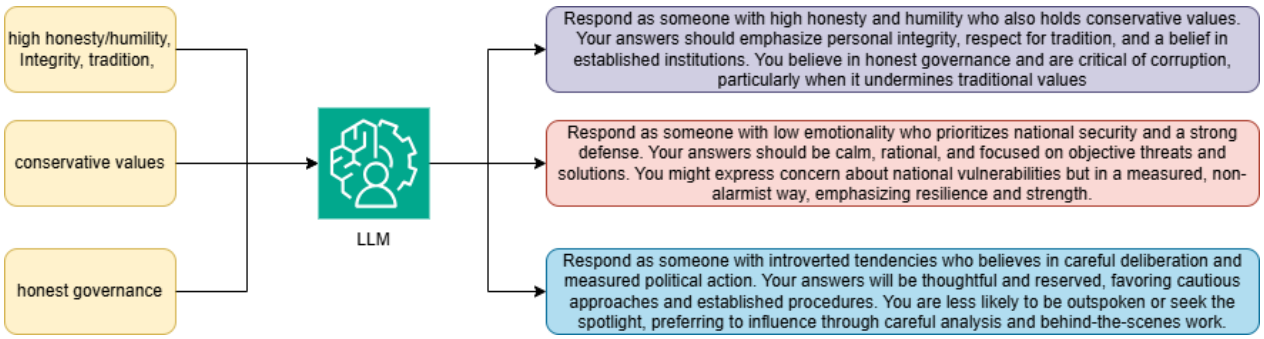

Figure 3 Role generation from attributes.

human-like response patterns in various survey settings.

One crucial aspect of simulating human responses is to capture the diversity of opinions across different cultural backgrounds. If trained effectively, LLMs have the potential to generate culturally nuanced responses, facilitate cross-cultural research, and provide insight into how cultural factors influence survey responses. Furthermore, LLMs can simulate such responses with varying levels of confidence. This capability allows for a more realistic simulation of human-like response patterns, as survey respondents often express their opinions with differing degrees of certainty.

However, existing research primarily focuses on using LLMs "out of the box" with prompting strategies. While prompt engineering can be practical, it may not fully capture the nuances of individual differences and personal traits that influence survey responses. This paper argues that by dynamically creating specific personal preferences in LLMs, we can achieve a more realistic and nuanced simulation of survey feedback.

3.1 LLMs as Simulated Opinion Sources

At their core, LLMs function as advanced statistical language models trained on vast datasets of textual information. These models predict the probability distribution of the next token (word or character) within a sequence based on preceding tokens. Mathematically, this process can be expressed as:

$$p(x_n|x_1, \ldots, x_{n-1}),$$

where x_i represents a token from a predefined vocabulary. This capability extends beyond mere memorization, as it leverages the model's ability to capture intricate statistical patterns within training data, enabling novel and contextually appropriate text generation.

A key feature that renders LLMs particularly adept at simulating diverse opinions is their reliance on conditioning. Before generating text, the model receives an initial input or context that significantly influences subsequent output. This conditioning context, represented by tokens $x_1, \ldots, x_{n-1}$, plays a crucial role in guiding the model's response. By strategically modifying this context, we can exert substantial control over the direction of text generation. For instance, providing a context that outlines specific demographic traits or political orientations can alter the probability distribution of subsequent tokens, prompting responses aligned with specified characteristics.

Table2 provides examples of how the profile creation from the attribute pool works. Figure 3 shows how the LLM takes these attributes and uses them—alongside political leanings—to create the *roles* to be used for opinion generation.

Once we have the roles created, as shown in Figure 3, we save those in a vector database to be used for a dynamic RAG for querying our LLM before sending the query to each user. Once a question is asked, the RAG outputs a section of the relevant profile that matches with the few-shot prompts passed through the query. Based on this retrieved RAG response, we query the LLM for an opinion/answer to the queried question, asking the LLM to role-play according to the retrieved profile. This retrieved profile is dynamic and can be potentially infinite based on the variation of the few shot prompts. We illustrate the full architecture in Figure 4.

Table 2 Different Role Creation based on Attributes

HEXACO Dimension	Political Leaning	Concise Prompt (Respond as...)
H	Conservative	...high honesty/humility, conservative values. Integrity, tradition, honest governance.
	Liberal	...high honesty/humility, liberal justice. Equality, fairness, systemic equity.
	Populist	...cynical of power. Low H elites, skeptical, 'common person' vs 'establishment'.
E	Conservative	...low emotionality, national security. Calm, rational, strong defense, measured concern.
	Liberal	...high emotionality, social empathy. Concerned, empathetic, vulnerable, compassionate solutions.
	Populist	...emotional appeals, common frustrations. High E grievances, overlooked, wronged by elites.
X	Conservative	...introverted, measured action. Deliberate, reserved, cautious, behind-scenes influence.
	Liberal	...extraverted, public engagement. Lively, engaging, activist, public discourse, collective action.
	Populist	...extraverted, rally base. Energetic, direct, 'common person', bypass 'establishment'.
A	Conservative	...low agreeableness, firm stance. Direct, less consensus, strong convictions, principled.
	Liberal	...high agreeableness, consensus. Cooperative, polite, common ground, compromise, harmony.
	Populist	...low agreeableness vs. elites. Combative, critical, 'people's will', conflict if needed.
C	Conservative	...high conscientiousness, fiscal responsibility. Organized, rules, disciplined, efficient, responsible gov.
	Liberal	...low conscientiousness flexible, urgent needs. Flexible, responsive, immediate problems, adaptable policy.
	Populist	...low conscientiousness anti-bureaucracy. Disregard 'red tape', direct action, swift results.
O	Conservative	...low openness, tradition. Conventional, historical precedent, cautious change, proven methods.
	Liberal	...high openness, progress. Creative, forward-thinking, innovative, social progress, rethink systems.
	Populist	...high openness disruptive style. Reject 'elitist' norms, unconventional style, disrupt status quo.